

Quax board - Sprint 4

Features we added (Sprint 4 Roadmap):

- **SR6:** Detect when BLACK forms a connected chain from top to bottom and display “BLACK wins.”
- **SR7:** Detect when WHITE forms a connected chain from left to right and display “WHITE wins.”
- **SR8.1:** Support Human V Bot mode where the bot automatically makes a valid move on its turn.
- **SR8.2 / SR8.3:** Provide a “Show Strategy” option for testing that displays a visual explanation of the bot’s chosen strategy.
- **SR8.4:** Provide a “Hide Strategy” option for testing that turns off the strategy display.

Testing we added (at least one per feature)

Feature	Test case
SR6	Completing a top-to-bottom BLACK chain sets BLACK as the winner and ends the game.
SR7	Completing a left-to-right WHITE chain sets WHITE as the winner and ends the game.
SR8.1	After the human makes a move, the bot automatically makes a valid move on its turn.
SR8.2 / SR8.3	Clicking “Show Strategy” displays the bot’s chosen strategy and shows the strategy highlight on the board.
SR8.4	Clicking “Hide Strategy” clears the strategy display and removes the strategy highlight.

Deviations / Notes

- Sprint 4 testing was split between game logic tests and UI tests, since win detection is game logic while the bot move and strategy display are UI behaviours.
- The current implementation runs in Human V Bot mode only; the mode selector is hidden and the title is set to “Quax - Human vs Bot”.
- The strategy display is a development/testing feature. It shows the bot’s selected move title/description and highlights the chosen board cell.
- The turn indicator now shows “Human” and “Bot” rather than player numbers, for example “BLACK (Human) to play”.